FABRICATION OF ECO FRIENDLY COOLING COMPARTMENT THROUGH PELTIER PRINCIPLE

PROJECT REPORT

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In partial fulfilment for the award of the course

of

GEA1401 - ENVIRONMENTAL SCIENCE (TCPR)

IN DEPARTMENT OF

MECHANICAL ENGINEERING

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
SAMAYAPURAM,TRICHY

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NOV 2026

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Under the Guidance of

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BONAFIDE CERTIFICATE

Certified that this project report titled **"FABRICATION OF ECO FRIENDLYCOOLING COMPARTMENT THROUGH PELTIER PRINCIPLE"** is the Bonafide work of **MANOJ S (8115U23ME025), SAMIDURAI V (8115U23ME037), SURAJ GAIRAV R S (8115U23ME051), and BARATH G (8115U23ME302).** who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported here in does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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Under
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DECLARATION BY THE CANDIDATES

We jointly declare that the project report entitled **"FABRICATION OF AN ECO-FRIENDLY COOLING COMPARTMENT THROUGH THE PELTIER PRINCIPLE"** is the result of the original work carried out by us. To the best of our knowledge, this work has not been submitted, either in whole or in part, to **Anna University, Chennai**, or any other university or institution for the award of any degree or diploma. This project report is submitted in partial fulfilment of the requirements for the award of the degree of **Bachelor of Engineering**.

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ACKNOWLEDGEMENT

It is with great pride that we express our gratitude and indebtedness to our institution,"**K.RAMAKRISHNAN COLLEGE OF ENGINEERING (AUTONOMOUS)**", for providing us with the opportunity to do this project.

We extend our sincere acknowledgment and appreciation to the esteemed and honorable Chairman, **Dr. K. RAMAKRISHNAN, B.E.,** for having provided the facilities during the course of our study in college.

We would like to express our sincere thanks to our beloved Executive Director, **Dr. S. KUPPUSAMY, MBA, Ph.D.,** for forwarding our project and offering an adequate duration to complete it.

We would like to thank **Dr. D. SRINIVASAN, B.E, M.E., Ph.D.,** Principal, who gave the opportunity to frame the project to full satisfaction.

We thank **Dr. H. RAMAKRISHNAN, M.E., Ph.D.,** Head of the Department, Mechanical Engineering, for providing his encouragement in pursuing this project. We wish to convey our profound and heartfelt gratitude to our esteemed project guide, **Mr. B. PITCHIA KRISHNAN, M.E, (Ph.D).,** Assistant Professor Department of Mechanical Engineering, for his incalculable suggestions, creativity, assistance and patience, which motivated us to carry out this project.

We render our sincere thanks to the Course Coordinator **Dr.M.CHRISPIN DAS M.E.,Ph.D Professor** and other staff members for providing valuable information during the course.

We wish to express our special thanks to the officials and Lab Technicians of our departments who rendered their help during the period of the work progress.

DEPARTMENT OF MECHANICAL ENGINEERING
Program: Bachelor of Engineering

VISION & MISSION OF THE COLLEGE

VISION:

- To achieve a prominent position among the top technical institutions.

MISSION:

- To bestow standard technical education par excellence through state-of-the-art infrastructure, competent faculty and high ethical standards.
- To provide education that can be used as a tool to transform the students.

VISION & MISSION OF THE DEPARTMENT

VISION:

- To provide eminent Mechanical Engineers to the society and industries through quality technical education and research.

MISSION:

- To impart quality professional education through state-of-the-art-infrastructure and resources.
- To motivate students and faculty to undertake research and development activities in the thrust areas of mechanical engineering.
- To carryout collaborative projects with industries and academics with fullest potential.

DEPARTMENT OF MECHANICAL ENGINEERING

Program: Bachelor of Engineering

Program Educational Objectives :(PEOs)

Graduates of Mechanical Engineering shall:

1. Have Successful careers in industries or opt for higher studies and research.

2. Analyze, synthesize and design mechanical and allied products addressing societal issues.

3. Exhibit technical expertise, communication skills, team work with professionalism with ethical values.

Program Specific Outcomes (PSOs)

Students shall:

PSO1: Identify, formulate and solve engineering problems in three core streams of Mechanical Engineering, i.e., Design, Thermal and Manufacturing Engineering

PSO2: Design, develop, test and implement energy efficient systems for required Engineering applications.

PROGRAM OUTCOMES

PROGRAM OUTCOMES (PO's)	
PO 1	**Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems
PO 2	**Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO 3	**Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO 4	**Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis an interpretation of data, and synthesis of the information to provide vali conclusions.
PO 5	**Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO 6	**The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice
PO 7	**Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO 8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO 9	**Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO 11	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PROGRAM's OUTCOME MAPPING

Check	PO number	Explanation
☑	**PO1 – Engineering Knowledge**	Applied heat transfer, airflow, and mechanical design principles to develop the grain dryer.
☑	**PO2 – Problem Analysis**	Identified grain-drying challenges and considered ways to reduce moisture and post-harvest losses.
☑	**PO3 – Design/Development of Solutions**	Designed a dryer to meet grain-processing needs while considering safety and usability.
☑	**PO4 – Conduct Investigations**	Tested the dryer and evaluated temperature, drying time, and grain condition.
☑	**PO5 – Modern Tool Usage**	Used suitable design, fabrication, and monitoring tools during project development.
☑	**PO6 – Engineer and Society**	Considered farmers' needs and the safe use of grain-drying equipment.
☑	**PO7 – Environment and Sustainability**	Aimed to reduce grain wastage and support sustainable use of harvested produce.
☑	**PO8 – Ethics**	Followed responsible engineering practices and reported project results honestly.
☑	**PO9 – Individual and Team Work**	Contributed to the team's design, fabrication, and project activities.
☑	**PO10 – Communication**	Presented the design, working method, and results through reports and presentations.
☑	**PO11 – Project Management and Finance**	Planned project tasks, selected components, and considered the project budget.
☑	**PO12 – Life-long Learning**	Learned about grain-drying technology and considered opportunities for future improvement.

SUSTAINABLE DEVELOPMENT GOAL



PRIMARY SDG's USED IN THE PROJECT



SDG 13: Climate Action: SDG 13 focuses on taking urgent action to combat climate change and its impacts by reducing greenhouse gas emissions. The proposed compost bin directly mitigates climate change by diverting organic waste from landfills, preventing the anaerobic decomposition that generates harmful methane gas. By converting waste into nutrient-rich compost locally, it reduces the carbon footprint associated with waste transportation and lowers reliance on chemical fertilizers.

SECONDARY SDG's USED IN THE PROJECT



SDG 11: Sustainable Cities and Communities: SDG 11 aims to make communities more sustainable. The proposed compost bin helps manage biodegradable waste through mechanical blending and compost conditioning, supporting better waste-management practices in households, hostels, and educational institutions.

ABSTRACT

Preserving perishable goods requires storage below ambient temperature. Conventional vapour-compression refrigerators do this, but they consume high electrical power, use refrigerants, have moving parts, and are bulky, which makes them unsuitable for compact and portable applications. This project deals with the design and fabrication of an eco-friendly, dual-mode (cooling and heating) compartment based on the Peltier principle.

The Peltier effect is the thermoelectric phenomenon in which a DC current passed through a junction of two dissimilar semiconductors (n-type and p-type) produces a temperature difference across the device. One face absorbs heat and the other rejects it. Reversing the current polarity reverses this action, so a single thermoelectric cooler (TEC) module can both cool and heat the compartment. The module is a solid-state heat pump with no refrigerant and no moving parts.The system consists of a TEC module, an LM35 precision temperature sensor, an AT89C51 microcontroller, an LCD, a regulated DC power supply unit (step-down transformer, bridge rectifier, filter and voltage regulator), a heat sink with cooling fan, and an insulated cooling cabinet. The LM35 senses the cabinet temperature, which is directly proportional to its output voltage. The AT89C51 compares the reading with the preset limit and switches the TEC in the required mode. The current temperature is displayed on the LCD. The cooling fan removes heat from the hot side of the module, and the cooling effect is conducted directly into the storage area.The system is compact, quiet, and environmentally friendly. It needs little maintenance and responds quickly to temperature changes. Its limitations are a limited heat-dissipation capacity and the trade-off between cost and power efficiency, so it suits small storage volumes. Possible applications include portable and car coolers, medical and vaccine storage, laboratory equipment, and small domestic coolers.

Keywords: Peltier effect, thermoelectric cooler, dual-mode heating and cooling, AT89C51, LM35, eco- friendly refrigeration

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LIST OF ABBREVIATIONS

S.NO	ABBREVIATION	WORD
1	AC	ALTERNATING CURRENT
2	ADC	ANALOGUE-TO-DIGITAL CONVERTER
3	CMOS	COMPLEMENTARY METAL-OXIDE SEMICONDUCTOR
4	COP	COEFFICIENT OF PERFORMANCE
5	DC	DIRECT CURRENT
6	I/O	INPUT / OUTPUT
7	LCD	LIQUID CRYSTAL DISPLAY
8	LM35	PRECISION CENTIGRADE TEMPERATURE SENSOR
9	PCB	PRINTED CIRCUIT BOARD
10	PID	PROPORTIONAL-INTEGRAL-DERIVATIVE
11	PWM	PULSE WIDTH MODULATION
12	TEC	THERMOELECTRIC COOLER
13	ZT	DIMENSIONLESS FIGURE OF MERIT

CHAPTER-1

INTRODUCTION

1.1 General Introduction

Many everyday goods, such as medicines, food, beverages and electronic components, must be kept below ambient temperature to remain usable. Conventional refrigeration meets this need with a compressor, a refrigerant and a condenser. Such systems are effective at large scale, but they are bulky, draw a high starting current, contain moving parts and rely on refrigerants that can harm the environment.

Thermoelectric cooling is a solid-state alternative. It uses the Peltier effect: when direct current flows through a junction of two dissimilar semiconductors, heat is absorbed at one face and released at the other. A Peltier cooler, also called a thermoelectric cooler (TEC), solid-state refrigerator or thermoelectric heat pump, contains no compressor, no refrigerant and no moving parts. Because the direction of heat flow depends on the direction of current, the same device can cool or heat. Although it can do both, in practice its main use is cooling.

The project, titled "Fabrication of Eco-Friendly Cooling Compartment through Peltier Principle", builds a compact, temperature-controlled compartment around a TEC module. The main elements are the TEC, an LM35 temperature sensor, an AT89C51 microcontroller, an LCD, a regulated power supply, a cooling fan with heat sink, and an insulated cooling cabinet. The sensor reports the compartment temperature to the microcontroller, which switches the TEC and fan to hold the set temperature. The same unit can also be used in heating mode by reversing the current.

Compared with a vapour-compression refrigerator, a thermoelectric device has a very different set of strengths. It has no compressor, condenser or refrigerant circuit, so it can be built into a very small enclosure and operated in any orientation. It runs from low-voltage DC, which makes it suitable for vehicles, battery-powered equipment and portable boxes. Its cooling effect begins as soon as current flows, and the cooling rate can be adjusted by changing the current. These features make the thermoelectric module attractive wherever a small heat load has to be removed quietly and with precise temperature control.

The performance of a thermoelectric cooler is, however, limited by the properties of the semiconductor materials and by how effectively heat is removed from the hot face. A module

pumps heat from the cold side to the hot side, and the electrical power supplied to the module is added to the heat that must be rejected. If the hot-side heat sink is undersized, the hot face warms up, the temperature difference across the module increases and the cooling capacity falls. Effective design therefore depends on a combination of module selection, heat-sink size, airflow, insulation of the cabinet and temperature control, rather than on the module alone.

Temperature control is the third essential element of the system. Perishable goods and sensitive items must be held within a defined temperature band, and a module that runs continuously wastes energy and may cool the cabinet below the required level. Using a sensor and a microcontroller, the module can be switched on and off automatically, so that the cabinet temperature remains near the set point. This project uses a low-cost LM35 sensor and an AT89C51 microcontroller, which are widely available and straightforward to program.

1.2 Need for Thermoelectric Cooling

Perishable goods must be stored below room temperature, which normally calls for a conventional refrigerator. For small loads, such as a few vaccine vials, a drink or a protected electronic enclosure, a full refrigerator is oversized, heavy and costly to run. A thermoelectric unit is compact, silent, responds quickly and can be controlled precisely, which makes it suitable for small cooling duties.

The need is sharpest in situations where a conventional refrigerator is impractical. Examples include carrying medicines between locations, keeping drinks or food cool in a vehicle, protecting temperature-sensitive electronic equipment in a closed enclosure, and cooling a small laboratory sample. In these cases the load is small, power is limited and portability matters. A compressor system sized for such a load is either unavailable or too heavy and costly.

A thermoelectric compartment also addresses an energy concern. Conventional compressors draw a high current at start-up and are efficient only when sized and loaded appropriately. A thermoelectric module operates directly from a low-voltage DC supply and can be switched on only when the temperature exceeds the limit. Although its coefficient of performance is lower than that of a compressor, the absolute energy use for a small, well-insulated cabinet can be modest. The practical benefit of the proposed unit should be assessed through measured power consumption and cooling time.

1.3 Importance of Eco-Friendly Operation

The TEC uses no refrigerant, so there is no leakage of ozone-depleting or high-global-warming-potential gases. It has no compressor, so it produces little noise or vibration, and it needs little maintenance. Its heating and cooling functions are combined in one unit, which reduces the number of devices needed. Its main drawback is lower efficiency than a vapour-compression system, so its use should be limited to applications where compactness, simplicity and control matter more than efficiency.

Refrigerants used in conventional equipment have contributed to ozone depletion in the past, and many modern refrigerants still have a significant global-warming potential if they leak. A thermoelectric unit avoids this issue because its working medium is the flow of charge carriers in a solid. At the end of its life the unit contains no refrigerant that must be recovered, although the electronic components and semiconductor materials should still be disposed of responsibly.

The environmental advantage also depends on the electricity used. Because thermoelectric modules are less efficient than compressors, the system should be properly insulated, sized to the actual load and switched on only when required. Powering the unit from a solar panel and battery, as considered in the future scope, would reduce its footprint further. The term eco-friendly in this report therefore refers mainly to the absence of refrigerant and the low maintenance requirement, and any claim about energy efficiency must be supported by measurement.

1.4 Overview of the Proposed Project

The proposed system places a TEC module on an insulated cabinet. The cold face is coupled to the cabinet interior and the hot face is coupled to a finned heat sink with a fan. The LM35 senses the cabinet temperature, and the AT89C51 compares it with the set value. When the temperature exceeds the limit, the controller switches on the TEC and fan. When the set temperature is reached, they switch off. The LCD shows the live temperature and the operating mode.

The operator places the items in the cabinet, closes the door and sets the required temperature. The micro - controller. reads the LM35 output through an analogue-to-digital converter, compares it with the set point and switches the TEC and fan through a relay or driver stage. While the temperature remains above the set point, the module continues to pump heat

from the cabinet to the heat sink. When the set point is reached, the module is switched off, and it is switched on again after the temperature rises by a small margin.

A power supply unit consisting of a step-down transformer, bridge rectifier, filter capacitor and voltage regulator provides the DC supply. The cooling fan mounted on the heat sink increases heat rejection from the hot face. The LCD displays the temperature and the operating mode. The structure is intended to be compact enough to place on a table or in a vehicle.

1.5 Scope of the Project

The scope covers concept development, component selection, circuit design, fabrication of the cabinet, programming of the controller and performance testing. The initial evaluation focuses on cooling a fixed volume to a set temperature. The unit can also be tried for cooling an enclosed space, such as a parked-car cabin or a protected area, and for heating mode. Each application needs its own testing because heat load, volume and insulation differ.

The multi-purpose nature of the unit lies in its ability to cool, to heat and to protect a space. The same module may be used to cool stored goods, to warm items when the current is reversed, or to keep the interior of an enclosed area within an acceptable temperature range. Each use case needs its own assessment of volume, insulation, heat load and power supply.

The project does not attempt to replace a household refrigerator. Its aim is to demonstrate a compact thermoelectric system with automatic control and to evaluate its performance in terms of cooling time, minimum temperature, temperature stability and power consumption. The limitations of the technology, including limited cooling capacity and modest efficiency, are acknowledged and discussed throughout the report.

CHAPTER-2

LITERATURE REVIEW

[1] **Bradley et al. (1998):** Presented the principles of mechatronics, covering sensors, signal conditioning, microcontroller-based control and actuators. The text supports the sensor–controller–actuator arrangement used in this project. Outcome: Provided the system-level basis for designing a closed-loop temperature controller.

[2] **Bolton (1999):** Described mechatronic system elements such as sensors, data presentation systems, microprocessor control and interfacing. Outcome: Guided the selection and interfacing of the LM35, controller and LCD.

[3] **Necsulescu (2002):** Covered mechatronic system design, sensors and actuators, and control of electromechanical systems. Outcome: Supported the choice of an on-off control strategy and switching of loads from a low-power controller.

[4] **Deb (1999):** Discussed automation, sensors and control technology in industrial systems. Outcome: Reinforced the use of feedback sensing for automatic operation.

[5] **Dwivedi and Kumar (2006):** Provided practical electronics circuits and construction guidance. Outcome: Supported the design of the power supply and interfacing circuits.

[6] **DiSalvo (1999):** Reviewed thermoelectric cooling and power generation, including the materials and figure of merit that limit device performance. Outcome: Showed that TEC efficiency is limited by material properties, which explains why TECs suit small cooling loads.

[7] **Riffat and Ma (2003):** Reviewed present and potential applications of thermoelectrics, including cooling, refrigeration and heat recovery. Outcome: Established that thermoelectric devices are practical for compact, quiet and low-maintenance applications.

[8] **Zhao and Tan (2014):** Reviewed thermoelectric cooling materials, modelling and applications. Outcome: Highlighted that heat-sink design and heat rejection from the hot side strongly affect cooling performance.

[9] **Rowe (1995):** Edited the CRC Handbook of Thermoelectrics, which collects the theory, materials, module design and applications of thermoelectric devices in one reference. The handbook treats the Seebeck, Peltier and Thomson effects and the design of modules for cooling and power generation. It is useful for relating module parameters, such as Seebeck

coefficient, electrical resistance and thermal conductance, to the cooling capacity of the proposed TEC. Outcome: Provided the theoretical basis for the equations and design approach used in Chapter 3.

[10] **Goldsmid (2010):** Presented a textbook treatment of thermoelectric phenomena, the figure of merit and the principles of thermoelectric refrigeration and generation. The work relates material properties to the temperature difference and efficiency that a module can achieve. Outcome: Established that the maximum temperature difference and COP of a TEC are governed by the material figure of merit, which explains why practical modules have limited performance.

[11] **Tritt and Subramanian (2006):** Gave an overview of thermoelectric materials, phenomena and applications. The article explained how device efficiency depends on material properties and surveyed uses of thermoelectric cooling and power generation. Outcome: Showed that thermoelectric cooling is an established technology for small, specialised loads, which supports the choice of a compact cooling compartment.

[12] **Snyder and Toberer (2008):** Reviewed complex thermoelectric materials and the strategies used to improve the figure of merit by balancing electrical conductivity, Seebeck coefficient and thermal conductivity. Outcome: Indicated that gains in module performance come mainly from materials research, so that a system designer should concentrate on heat rejection, insulation and control, which are within the designer's influence.

[13] **Bell (2008):** Discussed cooling, heating, power generation and waste-heat recovery with thermoelectric systems. The article argued that practical performance depends on the heat exchangers and the overall system design as well as on the thermoelectric material. Outcome: Supported the emphasis placed on the hot-side heat sink and fan in the proposed unit.

[14] **Riffat and Qiu (2004):** Compared thermoelectric air conditioners with vapour-compression and absorption systems. The comparison showed that conventional systems generally achieve a higher coefficient of performance, while thermoelectric units offer compactness, silence and the absence of refrigerant. Outcome: Indicated that thermoelectric cooling is best used for small loads where its other advantages outweigh its lower efficiency.

[15] **Chein and Huang (2004):** Studied the application of thermoelectric coolers to electronic cooling with heat sinks. They examined how the thermal resistance of the heat sink and the supplied current affect cooling performance. Outcome: Demonstrated that heat-sink design and

operating current must be selected together, which guides the heat-sink and power-supply design in this project.

[16] **Lineykin and Ben-Yaakov (2007):** Developed an electrical equivalent-circuit model of a thermoelectric module based on manufacturer data, allowing its thermal and electrical behaviour to be simulated. Outcome: Provided a method for predicting module behaviour before fabrication and for comparing measured and simulated results.

[17] **Min and Rowe (2006):** Built and tested prototype thermoelectric domestic refrigerators and examined the effect of the heat-exchanger arrangement on performance. Outcome: Showed that prototype performance depends strongly on heat exchange at both faces and provided a basis for planning the performance measurements of the present prototype.

[18] **Abdul-Wahab et al. (2009):** Designed and tested a portable thermoelectric refrigerator powered by a solar photovoltaic supply. Outcome: Demonstrated that thermoelectric cooling can run from a small DC source, supporting the solar-assisted option in the future scope.

[19] **Enescu and Virjoghe (2014):** Reviewed the parameters that influence thermoelectric cooling performance, including supply current, temperature difference and module geometry. Outcome: Provided a framework for selecting the operating current and for judging the measured COP of the prototype.

[20] **Vián and Astrain (2009):** Developed a thermoelectric refrigerator that used two-phase thermosyphons to improve heat dissipation from the hot side. Outcome: Showed that better heat rejection raises performance, suggesting heat pipes or thermosyphons as a later improvement to the proposed unit.

CHAPTER-3

BACKGROUND STUDY

3.1 Thermoelectric Effects

Three related effects govern thermoelectric devices:

- **Seebeck effect:** A temperature difference across two dissimilar conductors produces a voltage. This is the basis of thermoelectric generation and of temperature measurement by thermocouples.
- **Peltier effect:** When current flows through a junction of two dissimilar materials, heat is absorbed or released at the junction, depending on the current direction. This is the basis of thermoelectric cooling.
- **Thomson effect:** Heat is absorbed or released when current flows along a conductor that has a temperature gradient.

3.2 Structure of a Thermoelectric Cooler

A TEC consists of many pairs of n-type and p-type semiconductor pellets, commonly bismuth telluride. The pellets are connected electrically in series and thermally in parallel, and they are sandwiched between two ceramic plates. When DC flows through the module, carriers move heat from one plate (cold face) to the other (hot face). Reversing the current swaps the hot and cold faces.

Inside the module, each pair of n-type and p-type pellets forms a thermocouple. The pellets are joined by thin copper strips that are soldered to the inner surfaces of the ceramic plates. Current enters the module, passes through all pellets in series and leaves at the other terminal. In the n-type pellets, electrons carry heat away from the cold junction, and in the p-type pellets, holes do the same. Because the pairs are connected thermally in parallel, their individual cooling effects add up, and a module with many pairs can pump a useful quantity of heat.

The ceramic plates provide electrical insulation and mechanical strength while still conducting heat well. For this reason, the module can be clamped directly between a cold plate and a heat sink without short-circuiting the electrical path. The module is fragile in tension and shear, so it should be clamped evenly with a thin layer of thermal paste and should not be bent or twisted.

3.3 Basic Performance Relations

The heat absorbed at the cold face is approximately

$$Q_c = \alpha \cdot I \cdot T_c - \frac{1}{2} \cdot I^2 \cdot R - K \cdot \Delta T$$

where α = Seebeck coefficient of the module (V/K), I = current (A), Tc = cold-face temperature (K), R = electrical resistance of the module (Ω), K = thermal conductance of the module (W/K) and$\Delta T = T_h - T_c(K)$

The first term is the Peltier cooling, the second is Joule heating (half reaches the cold side) and the third is heat conducted back from the hot side. The input power is P = V·I, and the coefficient of performance is COP = Qc / P. Heat rejected at the hot face is Qh = Qc + P, so the heat sink must dissipate both the heat pumped and the electrical input.

3.4 Factors Affecting Performance

- Supply current and voltage applied to the TEC.
- Temperature difference between hot and cold faces.
- Effectiveness of the heat sink and fan on the hot side.
- Thermal contact between the TEC faces and the sink or cabinet (thermal paste, clamping pressure).
- Insulation and air tightness of the cabinet.
- Heat load and volume of the compartment.
- Ambient temperature.

The hot side must be cooled well. If the heat sink is poor, the hot face warms up and the cold face loses its cooling capacity.

Heat transfer at the interfaces deserves particular attention. Any air gap between the module and the cold plate or heat sink acts as an insulating layer and reduces the heat that can be moved. A thin, uniform layer of thermal paste and even clamping pressure reduce this resistance. The heat sink fins should be arranged so that the fan can blow air through them without obstruction, and the warm exhaust air must not be drawn back into the intake.

The cabinet itself is a major factor. A well-insulated and sealed cabinet loses little heat from the surroundings, so the module has to remove only the heat of the stored items and the

small leakage through the walls. A poorly insulated cabinet may never reach the set temperature, however good the module is.

3.5 Heating Mode

Reversing the polarity of the supply swaps the roles of the faces, so the same module heats the cabinet. This can be arranged with a relay or an H-bridge driven by the controller. Heating mode works with a higher COP than cooling mode, because the heat delivered equals the heat pumped plus the electrical input.

In practice, polarity reversal is achieved with a double-pole double-throw relay or an H-bridge circuit. The controller selects the direction according to the chosen mode. A short pause should be introduced before reversing the current, so that the relay contacts do not switch under full load and the module is not subjected to abrupt thermal shock. In heating mode the cabinet-side face becomes the hot face, so the temperature limit of the cabinet materials and the stored items should be respected.

3.6 Temperature Control

The controller uses on-off (thermostat) control with a small hysteresis band so that the TEC does not switch rapidly around the set point. More advanced control, such as PWM or PID, can be added later.

Hysteresis is important in on-off control. Without it, small fluctuations in the sensor reading near the set point can cause the relay to switch on and off repeatedly, which wears the contacts and stresses the module. A band of about one degree around the set point is a typical starting value and can be adjusted during testing. A minimum off-time can also be programmed to protect the module and the relay.

3.7 Figure of Merit

The quality of a thermoelectric material is expressed by its figure of merit, defined as

$$Z = \frac{\alpha^2}{\rho\, k}$$

where α is the Seebeck coefficient, ρ is the electrical resistivity and k is the thermal conductivity of the material. The dimensionless figure of merit ZT is obtained by multiplying Z by the absolute temperature. A good thermoelectric material should have a high Seebeck coefficient, a low electrical resistivity and a low thermal conductivity. These requirements conflict, because

materials that conduct electricity well generally also conduct heat well, which is why practical modules have a limited maximum temperature difference and a modest COP.

For a designer, the figure of merit is fixed by the choice of module. What can be changed is the operating current, the temperature difference across the module and the quality of heat rejection. The module should therefore be operated at a current that gives a good balance between cooling capacity and COP, and not at the maximum current, where Joule heating reduces efficiency.

3.8 Heat Load of the Cabinet

The module must remove the heat that enters or is generated in the cabinet. The main contributions are conduction through the walls and door, the heat content of the stored items, heat from air leakage when the door is opened and the small heat released by any internal fan or electronics. The heat conducted through the walls is estimated by

$$Q_{wall} = U\,A\,(T_{amb} - T_{cab}), \qquad U = \frac{1}{\sum(L_i/k_i)}$$

where U is the overall heat-transfer coefficient, A is the wall area, L is the thickness of each layer and k is its thermal conductivity. The heat to be removed from the stored items to cool them from the initial temperature to the set point is

$$Q_{product} = m\,c_p\,(T_i - T_{set})$$

where m is the mass and cp the specific heat of the items. Dividing this energy by the net cooling power of the module, after subtracting the wall heat gain, gives an estimate of the time required to cool the load. The calculation should be repeated with the measured values of the prototype, and the estimated and measured cooling times should be compared in the results chapter.

3.9 Heat-Sink Selection

The hot-side heat sink must reject the heat absorbed from the cabinet and the electrical power supplied to the module. Its performance is described by its thermal resistance from the sink to the surrounding air, Rsa. The temperature of the hot face can then be estimated as

$$T_h = T_{amb} + (Q_c + P)\,(R_{sa} + R_{contact})$$

where Qc is the heat absorbed at the cold face, P is the electrical input power and Rcontact is the thermal resistance of the interface between the module and the sink. A forced-air heat sink with a low thermal resistance keeps the hot face closer to ambient temperature and allows the cold face to reach a lower temperature. The fan should be rated to provide enough airflow across the fins, and the sink should be large enough for the heat load.

3.10 Temperature Sensing and Signal Conversion

The LM35 delivers an output voltage of 10 mV for every degree Celsius, so that the output at 30 °C is 300 mV. This voltage is too small to be read directly by the AT89C51, which has no analogue input. An analogue-to-digital converter changes it into a binary number. If an 8-bit converter such as the ADC0804 is used with its reference input set to give a full-scale range of 2.56 V, each count corresponds to 10 mV and therefore to one degree Celsius, which simplifies the conversion in software. The micro - controller.then reads the converter output through one of its ports, converts it to a temperature value and sends it to the LCD. The accuracy of the measurement also depends on the placement of the sensor, which should represent the air or product temperature rather than the temperature of the cold plate.

CHAPTER-4

PROBLEM IDENTIFICATION

Highly perishable goods must be kept below room temperature, which normally means using a conventional refrigerator. Such a refrigerator draws a large current and is bulky. The following problems establish the need for the proposed system.

Perishable goods, medicines and sensitive equipment are affected by temperature in different ways. Small changes in storage temperature can alter the rate of spoilage or reduce the effectiveness of a product. For this reason, a storage system should be judged by how reliably it holds the required temperature, and not only by how low a temperature it can reach.

4.1 High Energy Consumption

Compressor-based systems have a high starting current and a large rated power, which is excessive for small cooling loads.

The high starting current is a particular concern in vehicles and battery-powered systems, where the supply is limited and sudden loads can cause voltage dips. In domestic and small commercial settings, an oversized compressor operating intermittently for a small load wastes energy and increases the electricity bill. A smaller, directly controlled cooling element avoids this problem.

4.2 Bulky Size

A compressor, condenser and evaporator make the unit large and heavy, so it is not easy to carry or fit in small spaces.

Size matters in vehicles, laboratories and portable applications, where space is limited and weight affects usability. A compact thermoelectric unit can be built into a small box, a drawer or a cabinet wall. The module itself is only a few millimetres thick, so most of the volume of the unit is taken up by the heat sink, fan and insulation.

4.3 Moving Parts and Maintenance

Compressors and refrigerant circuits contain moving parts, wear out and need periodic servicing.

Reducing the number of moving parts also improves reliability. A thermoelectric module has none, and the only moving part in the proposed unit is the cooling fan, which is inexpensive and easily replaced. Maintenance is therefore limited to cleaning the heat sink and checking the electrical connections and thermal interfaces.

4.4 Environmental Concern

Refrigerants can leak and contribute to ozone depletion and global warming.

Regulations on refrigerants are becoming stricter, and the cost of servicing and recovering refrigerant adds to the lifetime cost of conventional equipment. A system that avoids refrigerants reduces compliance requirements and removes the risk of leakage, which is an advantage in small, unattended or mobile installations.

4.5 Separate Heating and Cooling

Conventional systems cool or heat, but one device seldom does both. Thermoelectric elements offer a compact structure, fast response and integrated cooling and heating.

Many applications need both cooling and heating, for example a compartment that must keep items cool in summer and warm in winter. With conventional equipment this requires two separate devices or a reversible heat pump with a complex valve system. A thermoelectric module performs both functions by simply reversing the current, which reduces the number of components and the space required.

4.6 Problem Statement

Small perishable loads and enclosed spaces need cooling below ambient temperature, but conventional refrigeration is bulky, consumes high current and has moving parts. A compact, eco-friendly, solid-state cooling compartment with automatic temperature control is therefore needed.

proposed system therefore sets out to provide an insulated compartment cooled by a thermoelectric module, an automatic control loop based on an LM35 sensor and an AT89C51 microcontroller, a display that shows the temperature and mode, and the ability to heat as well as cool. The system is intended for small loads, and its limits will be stated clearly after testing.

CHAPTER-5

IMPACT OF THE PROBLEM

5.1 Spoilage of Perishable Goods

If goods are not kept below the required temperature, their quality declines and they may spoil, causing food waste and, for medicines, loss of potency.

Medicines and biological products are especially sensitive. Exposure to temperatures outside the recommended range can reduce potency without any visible change, so the product may appear usable while it is not. Reliable temperature control is therefore a safety requirement as well as a quality requirement.

5.2 Economic Loss

Spoiled goods are a direct financial loss. A large refrigerator also adds to electricity cost when only a small load needs cooling.

The cost of cooling equipment includes purchase, running cost and maintenance. For small users such as shops, clinics and households, the cost of a full refrigeration unit may be out of proportion to the value of the goods stored. A low-cost thermoelectric unit offers an affordable alternative, although its running cost per unit of cooling is higher and should be compared with the alternative.

5.3 Energy and Peak Load

High-current appliances increase demand on the supply and are unsuitable where power is limited, for example battery-powered or vehicle applications.

In a vehicle, a large additional electrical load shortens battery life and may affect starting. A small thermoelectric unit can be sized to match the available supply and switched off automatically by the controller when it is not needed. Control of this kind keeps the average power demand low.

5.4 Environmental Impact

Refrigerant leakage and the electricity generated for large cooling units both add to environmental load.

Equipment manufacture, transport, use and disposal all have an environmental cost. A system built from a small number of parts, such as the proposed unit, has a lower material content than a compressor system. Its overall impact will still depend on the electricity source and on how long the unit remains in service.

5.5 Heat in Enclosed Spaces

In summer, the interior of a parked car or a closed protected area can rise well above a safe or comfortable limit. This damages stored items and electronics and makes the space unusable until it is cooled.

Cooling a closed space brings its own challenge. The heat gain is continuous, the volume of air is large compared with that of a small cabinet, and the roof and windows of a vehicle admit solar radiation. A thermoelectric unit can reduce the temperature of a small protected area, but it cannot be expected to cool a whole passenger cabin quickly, and its use in that role should be treated as a limited aid and not a replacement for air conditioning.

CHAPTER-6

OBJECTIVES

The overall objective is to design, fabricate and test an eco-friendly cooling compartment that uses the Peltier principle and can also provide heating from the same unit.

The proposed machine combines a thermoelectric module, a heat sink with fan, an insulated cabinet, a sensor, a microcontroller, a display and a power supply. Its performance will be evaluated through measurements of cooling time, minimum temperature, temperature stability, heating performance and power consumption.

6.1 Primary Objectives

- To fabricate a compact, insulated compartment cooled by a thermoelectric module.
- To avoid refrigerants and moving compressor parts.
- To provide automatic temperature control with a sensor and micro - controller.
- To display the live temperature and operating mode on an LCD.
- To combine heating and cooling in a single unit.

The first objective is to build a compact, insulated compartment in which a thermoelectric module can cool the stored items without any refrigerant. The cabinet should be sealed well enough to limit heat gain, yet easy to open, load and clean. The cold face of the module should be coupled to the interior through a plate or fin arrangement that spreads the cooling effect.

The second objective is to provide automatic temperature control. The sensor and the micro controller.should hold the cabinet temperature within a band around the set point, and the display should show the actual temperature and the operating mode so that the user can confirm that the system is working.

The third objective is to combine cooling and heating in a single unit. By reversing the current through the module, the same hardware should be able to warm the cabinet. This feature extends the usefulness of the unit and demonstrates the reversible nature of the Peltier effect.

The last primary objective is to demonstrate an eco-friendly solution. The unit should avoid refrigerants and should have only the fan as a moving part. This reduces maintenance and environmental risk and makes the unit suitable for use in places where servicing is difficult.

6.2 Technical Objectives

- To select a suitable TEC module, heat sink and fan for the intended heat load.
- To interface the LM35 sensor with the AT89C51 micro-controller.
- To design a regulated DC power supply (transformer, rectifier, filter and regulator) for the TEC, fan and controller.
- To switch the TEC and fan from the controller through a relay or driver circuit.
- To measure cooling time, lowest temperature reached and power consumption.
- To verify that the temperature stays within the set band.

The first technical objective is to select a module, heat sink and fan that match the expected heat load. The selection should consider the cooling capacity of the module at the expected temperature difference, the heat that the sink must reject and the airflow of the fan. These choices will be checked against the estimates of cabinet heat load and heat-sink performance given in Chapter 3.

The second technical objective is to interface the LM35 sensor with the AT89C51 micro - controller. The analogue output of the sensor must be converted to digital form, read by the micro-controller and displayed on the LCD. The program should include a set-point comparison, hysteresis and a safe default behaviour if the sensor reading is invalid.

The third objective is to design a regulated DC power supply that provides the required voltage and current for the module, the fan, the control circuit and the display. The supply should be protected by a fuse and should have enough margin for the starting current of the fan and the switching of the relay.

The fourth objective is to switch the module and fan from the controller through a relay or driver circuit that is suitable for the load current. The controller must not carry the load current directly, and the relay contacts should be rated with a safety margin.

The final objective concerns performance evaluation. Testing should record the cabinet temperature against time, the lowest temperature reached, the time needed to reach the set point, the supply voltage and current and the power consumed.

CHAPTER-7

HARDWARE REQUIREMENTS

7.1 Thermoelectric Cooler (TEC)

The TEC consists of two different semiconductors, one n-type and one p-type, with different electron densities. They are placed thermally in parallel and electrically in series and joined by a thermally conducting plate on each side. When current flows, the face with the cooling plate absorbs heat, which is moved to the other face, where the heat sink is mounted. TECs are usually connected side by side and sandwiched between two ceramic plates.

The module is selected according to the required cooling capacity, the supply voltage and the available current. A larger module has a greater capacity but also draws more current and releases more heat at the hot face, so the heat sink and power supply must be sized accordingly. Modules of the common 12 V, 40 mm × 40 mm class are widely used in small coolers and draw several amperes at full rating, so the supply must be able to provide this current continuously.

The module should not be operated without a heat sink and fan, because the hot face heats up quickly and the module can be damaged. The supply polarity must also be checked, since the cold and hot faces depend on it. The wires should be rated for the current, and the module should be mounted so that the cold face is coupled to the cabinet and the hot face to the heat sink with thermal paste between them.

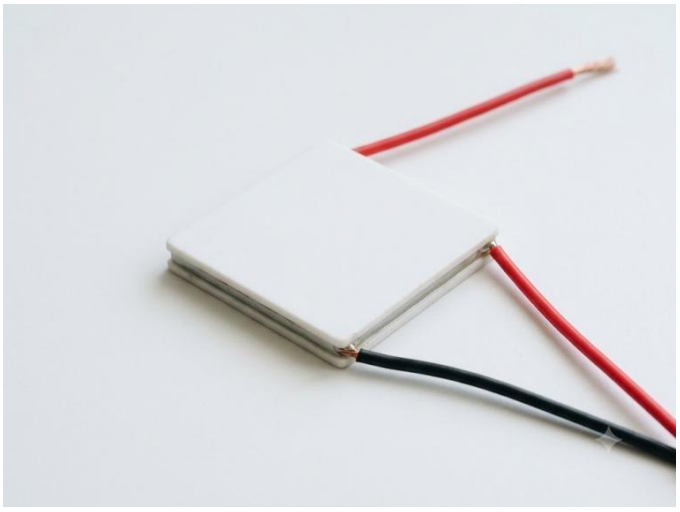


Fig 7.1.1 – Thermoelectric Cooler

7.2 LM35 Temperature Sensor

The LM35 series are precision integrated-circuit temperature sensors whose output voltage is linearly proportional to the Celsius temperature (10 mV/°C). It has an advantage over linear sensors calibrated in kelvin, because the user does not have to subtract a large constant voltage to obtain Celsius readings.

The LM35 has three terminals: supply, output and ground. It operates from a supply of 4 V to 30 V and draws only a very small current, so its own heating has a negligible effect on the reading in still air. Its low output impedance and linear response make it simple to connect to an analogue-to-digital converter without additional amplification in most cases. Long leads can pick up noise, so the sensor wires should be kept short or shielded, and a small capacitor or resistor network may be added at the output if the reading is unstable.

The sensor should be fixed inside the cabinet at a position that represents the air or product temperature. It should not touch the cold plate directly, because the plate is colder than the air in the cabinet and the reading would lead the controller to switch off too early.

Features:

- Calibrated directly in °Celsius.
- 0.5 °C accuracy guaranteeable (at +25 °C).
- Rated for the full −55 °C to +150 °C range.
- Suitable for remote applications.

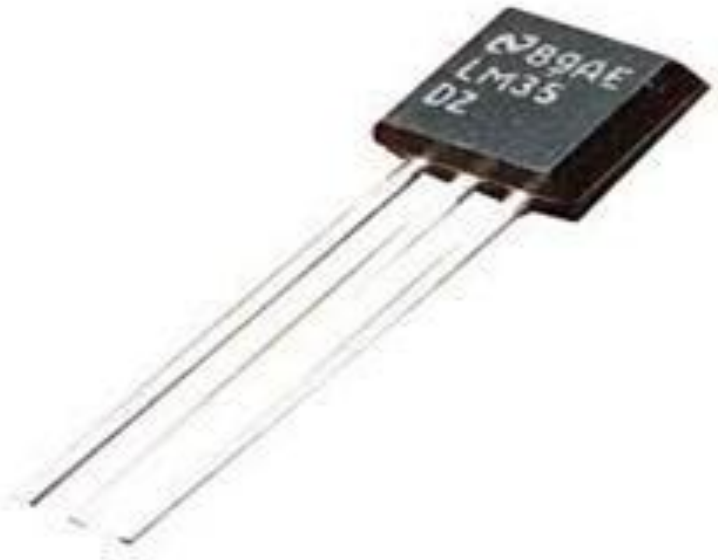


Fig 7.2.1 – LM35 Sensor

7.3 micro - controller.(AT89C51)

The AT89C51 is a low-power, high-performance CMOS 8-bit micro - controller.with Flash programmable and erasable read-only memory. It is manufactured using Atmel's high-density non-volatile memory technology and is compatible with the industry-standard 80C51 instruction set. Its standard features include 32 I/O lines, two 16-bit timer/counters, a full-duplex serial port, an on-chip oscillator and clock circuitry, and two software-selectable power-saving modes. Because the 8051 family has no built-in analogue-to-digital converter, the LM35 output is read through an external ADC (e.g. ADC0804). The micro - controller.requires a +5 V supply, a crystal oscillator with two small capacitors and a reset circuit that holds the device in reset for a short time after power-up. A crystal of 11.0592 MHz is commonly used because it allows standard serial baud rates and gives a convenient machine-cycle time. The ports are allocated to the different functions: one port reads the digital output of the converter, another carries the data lines of the LCD, and a few pins serve as control lines for the LCD and as outputs to the relay driver. The exact allocation depends on the circuit and should be shown in the circuit diagram.

The control program is written in assembly language or in embedded C. It initialises the ports and the LCD, reads the converter, converts the reading to a temperature, compares it with the set value and drives the outputs. It also includes a delay routine for the hysteresis and for the polarity-reversal pause. The program is loaded into the Flash memory of the device with a programmer.

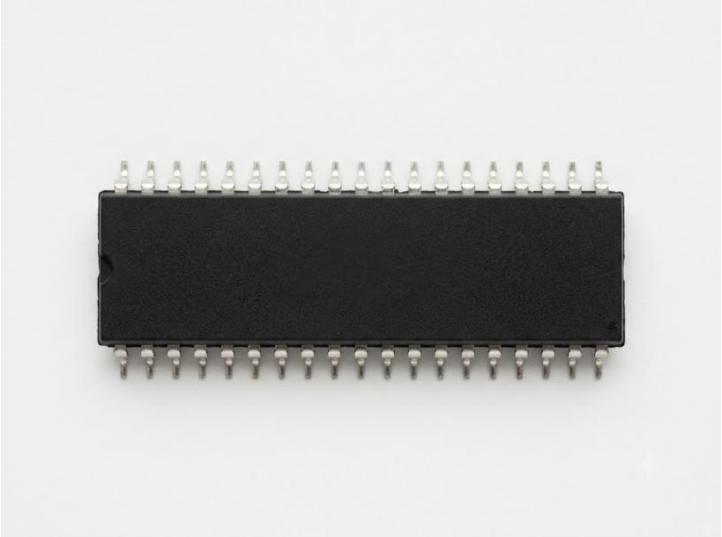


Fig 7.3.1 – Microcontroller

7.4 Power Supply Unit

Electronic equipment usually needs 6 V, 9 V or 12 V DC, which is much lower than the AC mains voltage. The power supply has four stages:

- **Transformer:** A step-down transformer reduces the mains voltage to the required level before rectification.
- **Rectifier:** Converts AC into pulsating DC. A bridge-type full-wave rectifier is preferred for its merits.
- **Filter:** Removes the ripple from the rectifier output and smooths the DC. The filter output stays constant only while the mains voltage and load stay constant, so a regulator is used at the output stage.
- **Regulator:** Eliminates ripple by setting the DC output to a fixed voltage.

A fixed regulator such as the 7805 provides the regulated +5 V needed by the microcontroller, the converter and the LCD. This regulator is suitable only for the low current of the control circuit. The thermoelectric module and the fan need a much larger current, so they are powered from a separate 12 V DC supply that has sufficient current capacity. Both supplies share a common ground. A fuse at the input protects the transformer and the circuit, and a heat sink on the regulator prevents it from overheating. The filter capacitor should be large enough to keep the ripple low at the maximum load current.

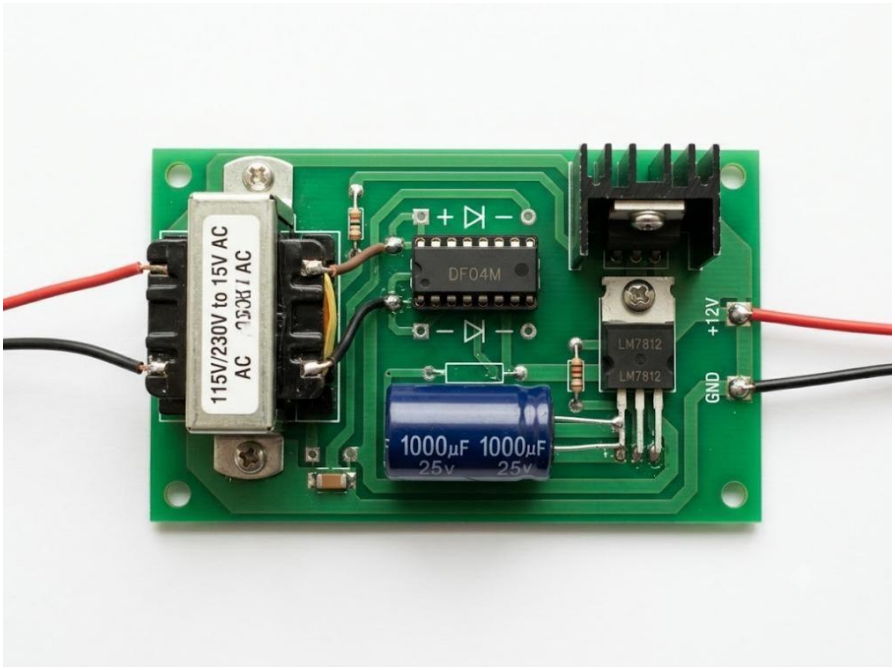


Fig 7.4.1 – Power Supply Unit

7.5 Liquid Crystal Display (LCD)

A liquid crystal display is a thin, flat panel that can either pass or block light. The panel is made up of blocks, each filled with liquid crystal that can be made clear or opaque by changing the electric current to that block. The LCD shows the cabinet temperature, set value and mode.

The display used in small projects is usually a 16 × 2 character LCD with a built-in controller. It can be connected to the micro - controller.using four or eight data lines and three control lines. A preset resistor adjusts the contrast. The first line can show the measured temperature and the second line the set point and operating mode. The display consumes little power, and its backlight can be switched off if the supply must be conserved.



Fig 7.5.1 – LCD

7.6 Cooling Cabinet

The cooling cabinet is the storage area of the TEC set-up. The cooling effect of the TEC is conducted directly to this area. It should be insulated and sealed to limit heat gain. [Add material, dimensions and volume.]

The cabinet walls are made from an insulating material such as expanded polystyrene or polyurethane foam, protected by an outer shell. The inner surface should be smooth and easy to clean. A metal cold plate, preferably aluminium, is mounted on the inner side of the module and may carry fins to spread the cooling into the air. The door must close tightly, and a rubber or foam gasket reduces air leakage. The size of the cabinet should be kept as small as the application permits, since a smaller volume needs less cooling power and gives a quicker response.



Fig 7.6.1 – Cooling Cabinet

7.7 Cooling Fan

A cooling fan is mounted close to the TEC for active cooling. Fans can draw cooler air in from outside, expel warm air from inside or move air across a heat sink to cool a particular component. Here it carries heat away from the hot-side heat sink. The fan should be chosen to give enough airflow through the heat-sink fins without excessive noise or current. A 12 V DC axial fan is commonly used. A second small fan can be placed inside the cabinet to circulate air across the cold plate and reduce temperature differences within the cabinet, if testing shows that this is useful. The fan should be switched on together with the module and may be kept running for a short time after the module switches off, so that the residual heat in the sink is removed.



Fig 7.7.1 – Cooling Fan

7.8 Switching Device

A relay or driver transistor connects the low-power controller to the TEC and fan, so that the controller does not carry the load current directly. The device must suit the load voltage and current.

The relay coil needs more current than a micro - controller.pin can supply, so a transistor or a driver chip is placed between the pin and the coil. A flyback diode across the coil protects the driver from the voltage spike produced when the coil is switched off. An optocoupler may be added to isolate the control circuit from the load. For reversing the polarity of the module, a double-pole double-throw relay or an H-bridge is used. The contact rating of the relay must exceed the maximum current of the module with a safety margin.

7.9 Heat Sink and Cold Plate

The heat sink on the hot face is a finned aluminium block that increases the surface area available for heat transfer to the air. Its thermal resistance should be low enough to keep the hot face near the ambient temperature at the full heat load. The cold plate on the cabinet side conducts heat from the air of the cabinet to the cold face of the module. Both surfaces must be flat so that they make good contact with the module, and both are coated with thermal paste before assembly.

The module is clamped between the plates using screws with insulating washers, and the screws are tightened evenly in steps. Excessive or uneven tightening can crack the ceramic plates, while weak clamping leaves air gaps that raise the thermal resistance.

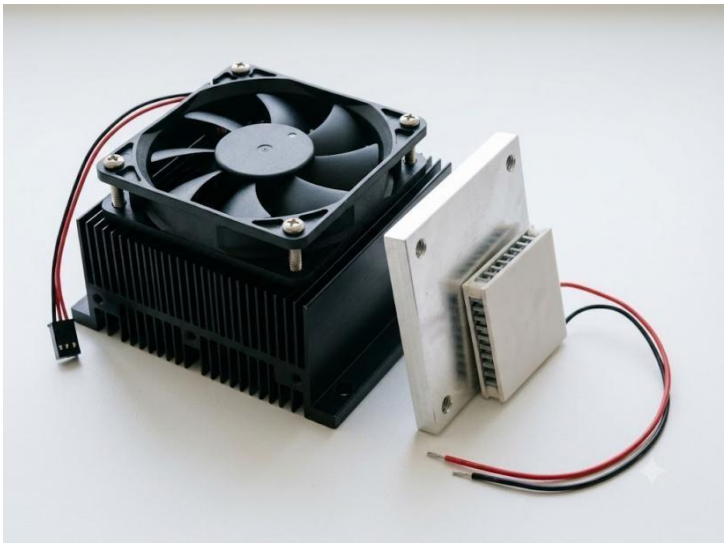


Fig 7.9.1 – Heat Sink and Cold Plate

7.10 Insulation

Insulation reduces heat gain into the cabinet through its walls and door. The thickness and conductivity of the insulating layer determine how much cooling power is wasted on heat leakage. Polyurethane foam has a low thermal conductivity and is easy to shape, while expanded polystyrene is cheaper and lighter. The insulation should be sealed on the outside, so that it does not absorb moisture, and gaps and joints should be filled to prevent thermal bridges.

Condensation can form on the cold plate in humid air, and the water should be drained or kept away from electrical parts. The wiring entry points should be sealed.



Fig 7.10.1 – Insulation Materials

7.11 Wiring and Protection

The wiring must carry the module current with little voltage drop and without overheating, so the cable cross-section should be chosen for the highest current. Connections should be soldered or fitted with proper terminals and covered with insulation. Fuses should be fitted in the mains input and the 12 V module supply. The mains-connected parts should be enclosed, and the earth connection should be provided where the case is metallic.

A thermal cut-off may be placed on the heat sink to disconnect the module if the fan fails and the sink overheats. This protects the module from damage due to excess hot-side temperature.

7.12 Supporting Structure

The supporting structure holds the cabinet, the heat sink and fan, the power supply and the control circuit. It should have enough strength to carry the weight and should hold the parts so that the airflow around the heat sink is not blocked. The electronic parts are placed in a separate enclosure away from the hot air outlet and are fixed so that vibration from the fan does not loosen the connections. Rubber feet or a base plate give the unit stability.



Fig 7.12.1 – Supporting Structure

CHAPTER-8

WORKING PRINCIPLE

8.1 Operating Principle

The system works on the Peltier effect. The DC power supply provides the power for the TEC. A Peltier cooler is a solid-state active heat pump that transfers heat from one side of the device to the other, using electrical energy, in a direction that depends on the current direction. It can also act as a temperature controller that either heats or cools.

Unlike a compressor system, the thermoelectric system has a simple heat path. Heat from the cabinet air is conducted through the cold plate to the cold face of the module, is pumped through the module together with the electrical input power and is released from the hot face into the heat sink. The fan then carries the heat from the fins into the surrounding air. Each part of this path adds thermal resistance, and the temperature difference across the module is the sum of the cabinet-to-ambient difference and the temperature drops across these resistances. Reducing any of the resistances improves cooling.

8.2 Operation

1. The mains supply is stepped down, rectified, filtered and regulated to give the DC supply for the TEC, fan, sensor and controller.
2. The LM35 senses the temperature inside the compartment and gives a voltage proportional to it.
3. The ADC converts this voltage to a digital value, which the AT89C51 reads and compares with the set temperature.
4. If the temperature is above the limit, the controller switches on the TEC and the fan. The cold face absorbs heat from the cabinet and the heat sink and fan release it to the surroundings.
5. When the temperature falls to the set value, the controller switches the TEC off.
6. The LCD shows the temperature and the operating mode throughout.
7. In heating mode, the TEC current is reversed so that the cabinet-side face becomes the hot face.

For example, when a vehicle is parked in summer and the interior temperature exceeds the limit, the TEC starts cooling the air to keep the interior within an acceptable temperature.

8.3 Limitation of the Principle

The main disadvantage of a Peltier cooler is that it cannot combine low cost with high power efficiency at the same time.

8.4 Preparation and Loading

Before operation, the cabinet is inspected for cleanliness, and the door seal and the cold plate are checked. The heat sink fins and fan are examined for dust or obstruction, and the wiring and connectors are checked for firm connection. The power supply is checked for the correct output voltages before the module is connected.

The items to be cooled are placed in the cabinet so that air can circulate around them. The cabinet should not be overloaded, because crowding blocks the air movement and slows cooling. The set temperature is selected, and the initial temperature of the cabinet and the ambient temperature are recorded. The door is then closed.

8.5 Sensing and Control Logic

The LM35 continuously produces a voltage proportional to the cabinet temperature. The converter changes this voltage to a digital number, and the micro - controller.reads it at regular intervals. The program converts the number to degrees Celsius and compares it with the set temperature. The control logic is as follows:

1. If the temperature is higher than the set value plus the hysteresis band, the controller switches on the fan and then the module in cooling mode.
2. If the temperature falls to the set value, the controller switches the module off. The fan may continue for a short time.
3. If the temperature rises above the upper limit again, the cycle is repeated.
4. If the sensor reading is outside the valid range, the controller switches the module off and shows a fault message.

The controller starts the fan before the module, so that the heat sink is cooled from the beginning. This prevents the hot face from heating up before airflow has been established.

8.6 Cooling Mode

In cooling mode, the current flows through the module in the direction that makes the cabinet-side face the cold face. The cold face absorbs heat from the cold plate, which in turn cools the air and the items in the cabinet. The temperature of the cabinet falls quickly at first,

because the temperature difference across the module is small, and more slowly as the difference grows and the heat leakage into the cabinet increases.

The lowest temperature that can be reached is determined by the balance between the cooling capacity of the module and the heat gain of the cabinet. If the lowest temperature is not low enough, the remedy is to improve insulation, increase the heat-sink performance or use a larger module, and not to increase the current beyond the rated limit.

8.7 Heating Mode

In heating mode, the controller reverses the current through the module by operating the polarity relay. The cabinet-side face becomes the hot face, and the module delivers to the cabinet both the heat pumped from the heat sink side and the electrical input power. The temperature control logic is the same as in cooling mode, but with the sense of the comparison reversed: the module is switched on when the temperature falls below the set value and off when it rises above it.

A short pause is inserted between the end of one mode and the start of the other, so that the relay does not switch under load. In heating mode the temperature must be limited to the safe value for the module, the cabinet materials and the stored items.

8.8 Heat Rejection

In cooling mode, the heat sink on the hot side rejects both the heat taken from the cabinet and the electrical power supplied to the module. The fan drives air through the fins to carry this heat away. If the airflow is blocked or the fan fails, the hot face temperature rises and the cooling capacity falls rapidly. The unit should therefore be placed where air can enter and leave freely, away from walls and other heat sources.

In heating mode, the heat sink side becomes the cold side and absorbs heat from the surrounding air. The fan helps by supplying air at ambient temperature to the sink.

8.9 Display and Operator Interface

The LCD shows the measured temperature, the set point and the operating mode, and any fault message. The operator can change the set point through push buttons connected to the microcontroller, and a mode switch selects cooling or heating. A status indicator shows when the module is on. The messages should be short and clear so that the operator understands the state of the unit at a glance.

8.10 Fault Handling and Shutdown

The controller should treat a sensor fault, such as an open or shorted sensor lead, as a reason to switch off the module and show a message. If the fan is not running, the module should not be switched on. A thermal cut-off on the heat sink provides an independent protection against overheating.

For shutdown, the module is switched off first and the fan is kept running for a short time to remove the residual heat from the sink. The supply is then disconnected. The operator records the final cabinet temperature, the time taken to reach the set point and the supply readings for each trial.

These steps provide a repeatable operating sequence for the cooling and heating of the compartment.

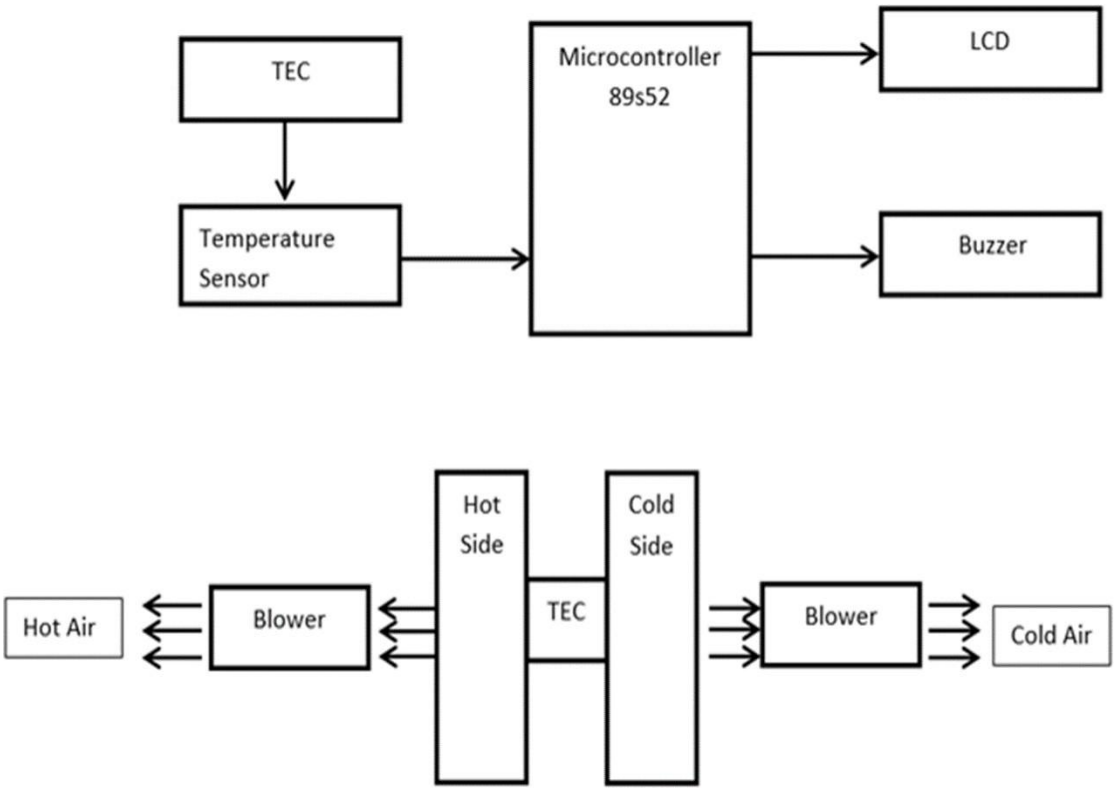


Fig 8.1 – Functional Block Diagram

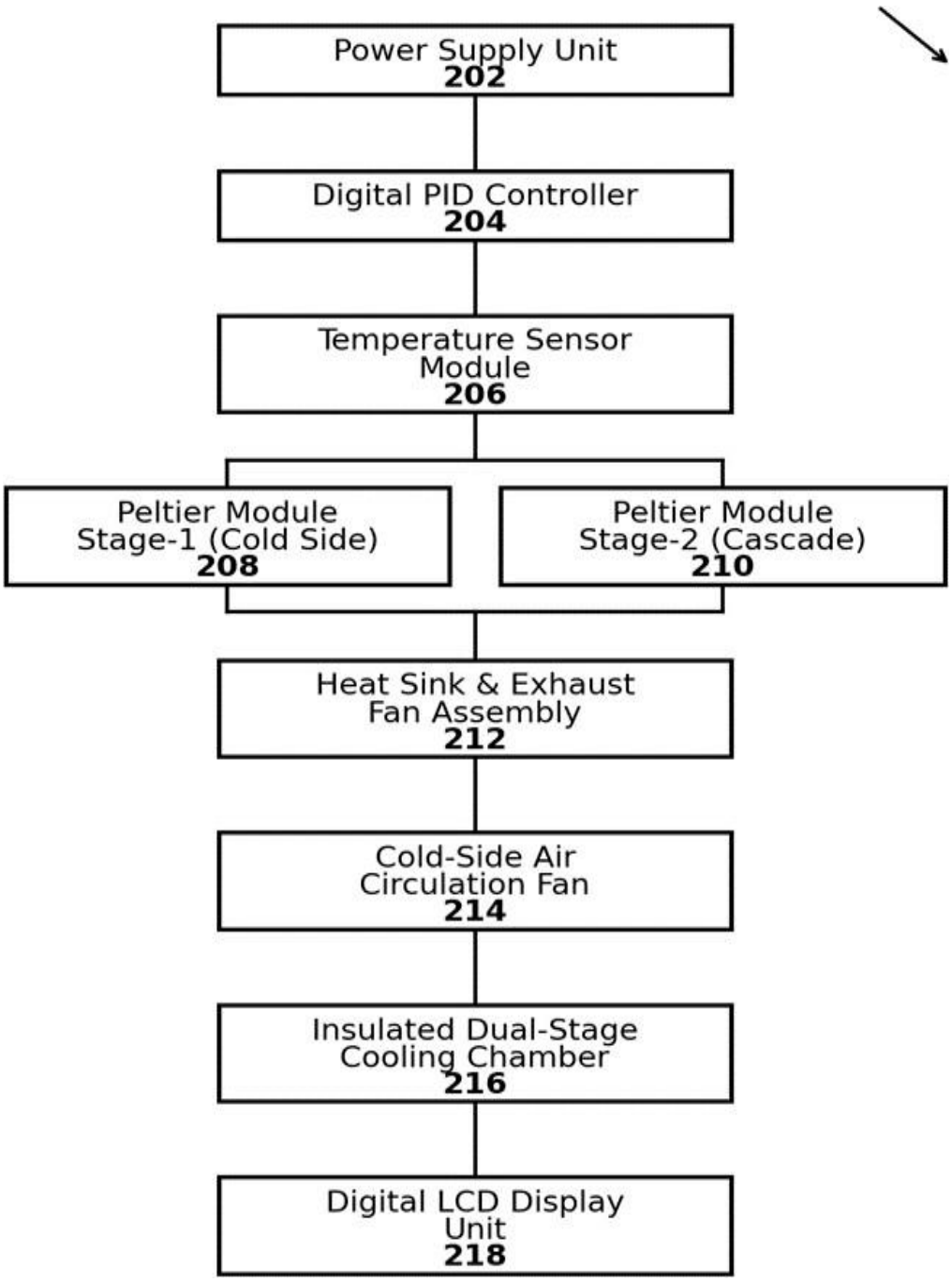


Fig 8.2 – Workflow / Flowchart

CHAPTER-9

2D DESIGN

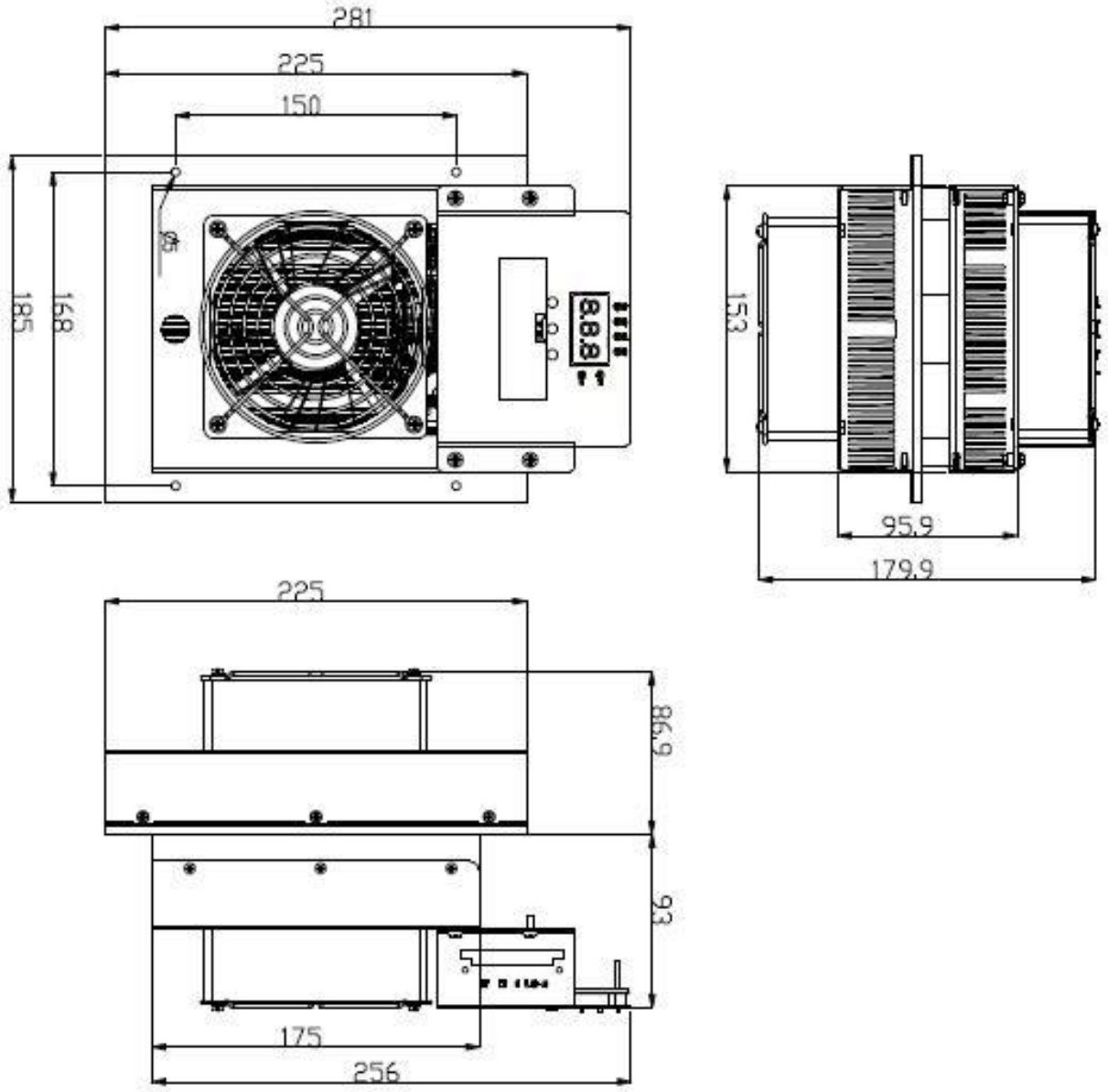


Fig 9.1 2D Design

CHAPTER-10

PROTOTYPE MODEL



Fig 10.1 Prototype Model Image 1



Fig 10.2 Prototype Model Image 2

CHAPTER-11

COST ESTIMATION

11.1 Bill of Materials

S.No.	Item	Specification / purpose	Qty
1	Thermoelectric cooler	Peltier module, cooling/heating	2
2	Heat sink and cooling fan	Hot-side heat rejection	1
3	LM35 sensor	Temperature sensing	2
4	AT89C51 microcontroller	Control unit	1
5	LCD	Display	1
6	Power supply unit	Transformer, rectifier, filter, regulator	12VDC
7	Relay / driver circuit	Switching of TEC and fan	1
8	Cooling cabinet with insulation	Storage area	2
9	Wiring, PCB and miscellaneous	Connections and fittings	1

Tab 11.1- Bill of Material

11.2 TABLE OF ESTIMATION

S.No.	Component	Estimated cost (₹)
1	TEC module	800
2	Heat sink and fan	1000
3	LM35 sensor	100
4	AT89C51 and ADC	500
5	LCD	200
6	Power supply unit	1000
7	Relay and driver	400
8	Cabinet and insulation	1200
9	Wiring and miscellaneous	800
	Total	**6000**

Tab 11.2 – Cost Estimation

CHAPTER-12

RESULT/OUTPUT

The prototype was tested for automatic cooling and heating of the compartment. The cabinet was kept at the ambient temperature and loaded with a small test load. The supply was then switched on, and the cabinet temperature was observed on the LCD until the set point was reached.

The TEC absorbed heat from the cabinet at its cold face, and the heat sink and fan removed it at the hot face. The LM35 sensed the cabinet temperature, and the AT89C51 compared it with the set value and switched the TEC and fan. The cabinet cooled to the set temperature within a few minutes for a small load. It was then held there automatically.

The controller switched the module on and off around the set point, so the cabinet temperature stayed within a small band. The width of the band depended on the hysteresis setting. A small hysteresis gave stable control without frequent switching of the relay.

In heating mode, reversing the current through the same module raised the cabinet temperature to the set value in a short time. Heating was faster than cooling, because the heat delivered includes the electrical input power. Using one module for both modes reduced the number of devices needed.

The hot side stayed close to the ambient temperature while the fan was running. This kept the temperature difference across the module low and showed that the heat sink and fan were suitable for the heat load. A larger heat sink or better airflow would improve the performance further.

The module drew power only while it was switched on, so the average power was lower than the rated power. The unit had no moving parts apart from the fan, so it ran quietly and needed little maintenance.

The cooling capacity was limited, and the lowest temperature reached depended on the ambient temperature. The unit therefore suits only small volumes and a limited heat load, and it cannot replace a conventional refrigerator.

CHAPTER-13

CONCLUSION

The project "Fabrication of Eco-Friendly Cooling Compartment through Peltier Principle" provides a compact, solid-state alternative to compressor refrigeration for small cooling duties. A thermoelectric module, supported by a heat sink and fan, cools an insulated cabinet. An LM35 sensor, AT89C51 micro - controller.and LCD provide automatic control and display. The absence of refrigerant and the near-absence of moving parts make the unit environmentally friendly and low in maintenance. Cooling and heating are available in one unit.

The main limitations are modest efficiency and limited heat-dissipation capacity, so the design is not suitable for large volumes. Many researchers and companies are working to make Peltier air-conditioning both cheap and efficient. With measured performance data, the prototype can be refined for practical use in small storage and enclosed-space cooling.

The design combines a few simple parts in a way that can be built, understood and repaired by a student or a small workshop. The module provides the cooling and heating effect, the sensor and micro - controller.give automatic control, and the heat sink, fan and insulation determine how well the effect can be used. The performance of the unit depends on all of these parts, and attention to heat rejection and insulation gives the greatest return.

The temperature control loop is simple but effective for this application. The use of an on-off controller with hysteresis protects the module and the relay, while the LCD gives the user the information needed to operate the unit. More advanced control can be added later without changing the hardware.

The project also shows how an engineering concept is turned into a working device: the literature is studied, the problem is defined, the components are selected using calculation, the prototype is built and its performance is tested. The test results should be used to refine the design, for example by changing the heat sink, the insulation or the control settings.

TEC modules are used in military equipment, aviation, aerospace, medical treatment, epidemic prevention, experimental apparatus and consumer products, such as water coolers, car coolers, hotel coolers, wine coolers, personal mini coolers and cool-and-heat sleep pads.

CHAPTER-14

FUTURE SCOPES

Enhanced Cooling Capacity and Performance: Develop higher-capacity cooling systems using multiple TEC modules and improved heat dissipation mechanisms.

Intelligent Temperature Regulation: Integrate PID-based control and advanced sensors to maintain precise temperature levels while minimizing energy consumption.

Renewable Energy Integration: Incorporate solar panels and battery storage to enable sustainable, off-grid cooling applications.

IoT-Based Smart Monitoring: Implement remote temperature monitoring, real-time alerts, and data logging for improved reliability and automation.

Advanced Thermal Insulation: Explore high-performance insulation materials to reduce heat transfer and improve overall energy efficiency.

Expansion into Practical Applications: Adapt the system for medicine storage, portable refrigeration, emergency cooling, and vehicle temperature management.

CHAPTER-15

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CHAPTER-16

APPENDIX

16.1 Evidence of Work



Fig 16.1 – Assembly of TEC and Heat Sink

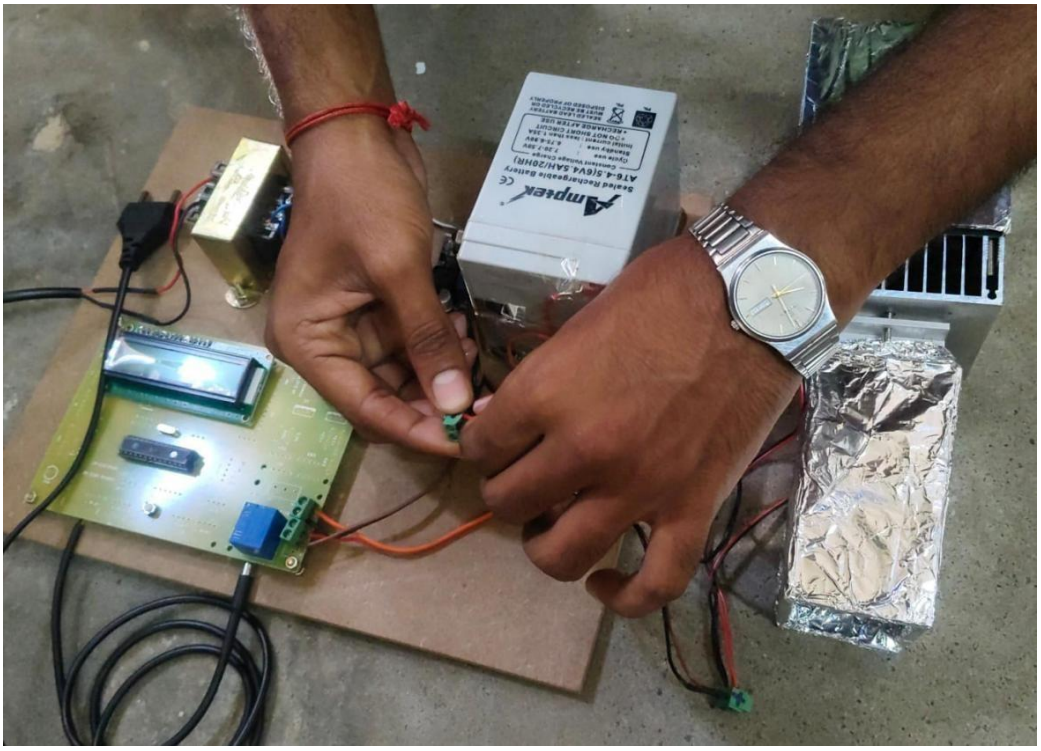


Fig 16.2 – Circuit Wiring

16.2 Project Gallery

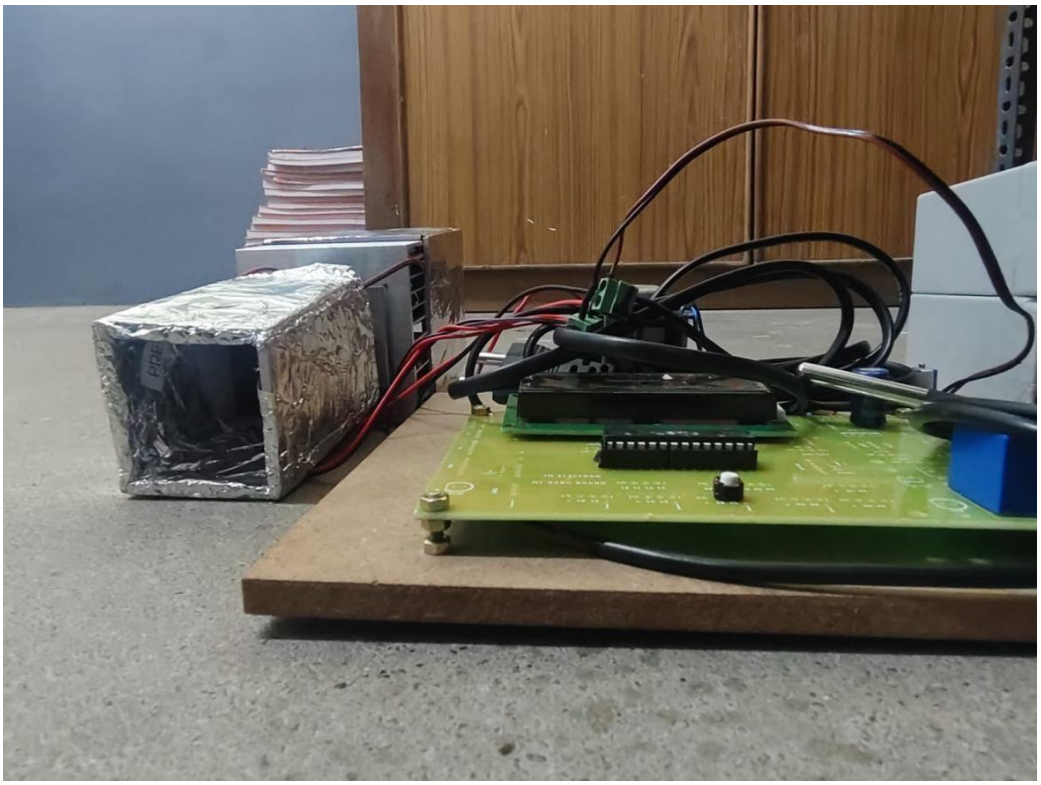


Fig 16.3 – Prototype Front View

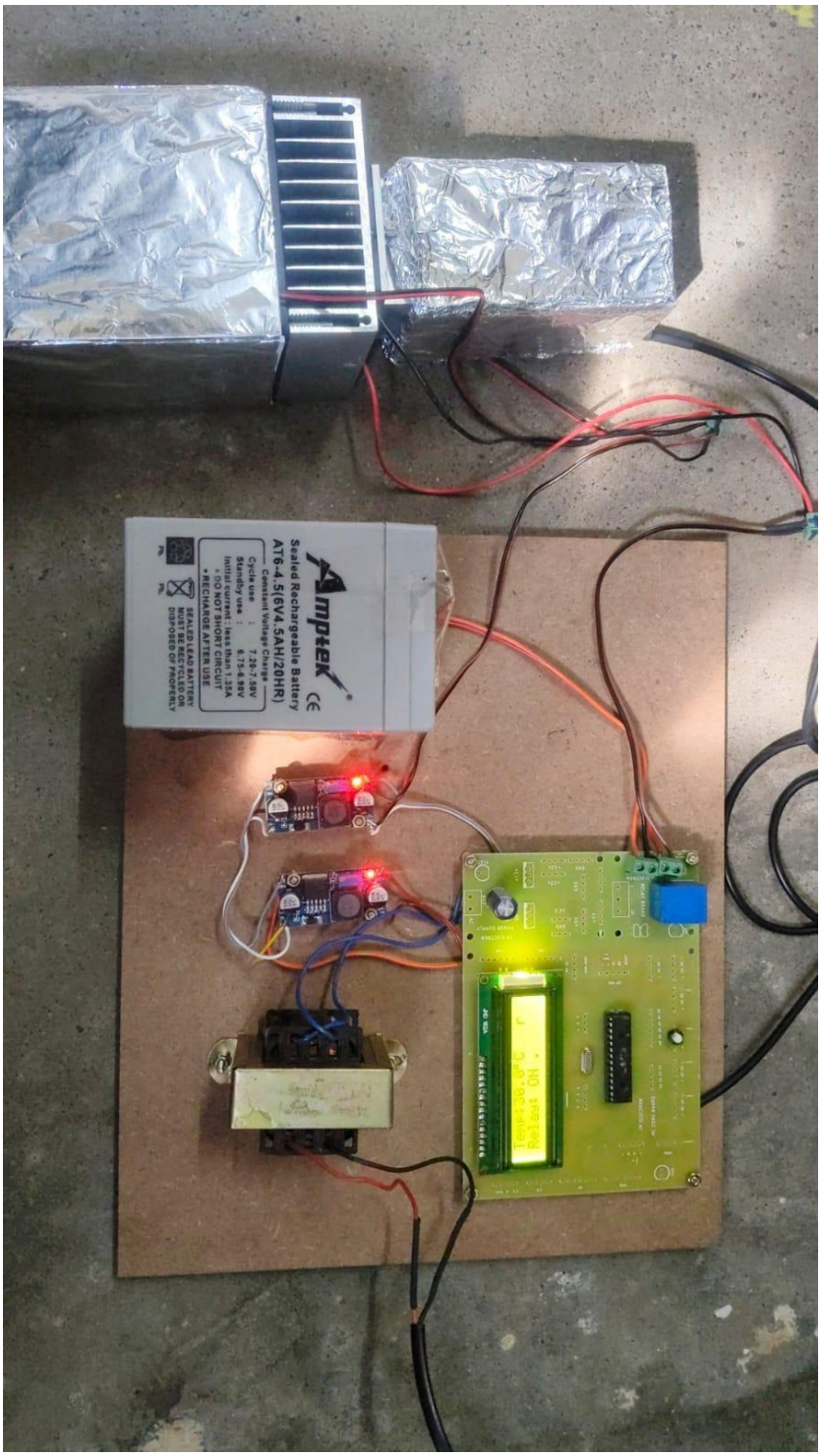


Fig 16.4 –Top View